

Food Safety Intelligence System

User Guide & Service Overview

How it works. What you receive. Quality assurance. Analytical methods. Industry landscape.

BUILT & WRITTEN BY

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1. What is FSIS?

The Food Safety Intelligence System (FSIS) is a fully automated, AI-powered global pathogen recall monitoring platform built and operated by Advanced Food-Tech Solutions (AFTS). It tracks regulatory recall notices from 70+ official sources across 60+ countries, filters for pathogen-related hazards, and delivers structured intelligence to food safety professionals.

FSIS is not a raw data feed. Every recall is enriched, normalized, severity-ranked, and interpreted through the lens of food process engineering. Weekly and monthly briefings read like a report from a practicing process authority — naming specific failure modes, referencing regulatory frameworks (21 CFR 113/114, PMO, HACCP), and committing to concrete engineering recommendations.

You get intelligence, not data. The analysis speaks the language of food engineers, not the language of databases.

1.1 What's included

Feature	What you get
Live dashboard	Interactive intelligence platform with pathogen filtering, live feed, severity tiers, outbreak flags, cross-region comparisons. Always current.
Daily briefs	Auto-generated HTML page each morning showing yesterday's recalls worldwide, with Tier-1 / outbreak triage and direct links to source regulatory notices.
Weekly digest	Every Friday: AI-written narrative with top 5 ranked threats, leading-pathogen analysis, regional breakdown. Delivered to your inbox.
Monthly report	9-section analyst-grade briefing: trend analysis, hotspot matrix, outbreak timeline, concentration metrics, predictive models, engineering recommendations. With Appendix listing every recall. PDF attached.
Yearly report	Annual pathogen surveillance review: year-over-year trend analysis, geographic shifts, emerging-threat forecast for the year ahead. Shipped at the close of each year.
Custom alerts	Build rules on Pathogen × Product × Country × Brand (AND/OR logic). Get an email within minutes of a matching recall appearing. 2 rules included.
XLSX dataset	Full recall history — every pathogen, every country, every source. Download, query, chart, build your own reports. Updated within minutes of every newly published recall.

2. How it works

FSIS operates a continuous automated pipeline that runs 24/7 with no manual intervention. The pipeline has four stages: **collection**, **enrichment**, **verification**, and **analysis & delivery**. Verification is the stage that distinguishes FSIS from a conventional data feed — every candidate recall passes through a dual-reviewer gate before it can reach the published dataset.

2.1 Collection

Automated scrapers connect to 70+ regulatory agency feeds worldwide. The primary scrape runs daily; targeted critical-agency sweeps run additionally during high-publication windows. Between these, four AI-powered gap-finders sweep for any recalls the structured scrapers might have missed, staggered across the day to catch postings from every time zone:

- **03:00 Athens** — overnight sweep (APAC + Oceania catch-up)
- **12:00 Athens** — midday worldwide sweep for missed recalls
- **14:00 Athens** — EU afternoon catch-up with Google-grounded regulator search
- **22:00 Athens** — US end-of-day sweep (15:00 ET / 12:00 PT)

Each gap-finder uses a distinct retrieval engine — Tavily, Exa, Google grounded search, and an aggregated cascade fallback — so a single index outage cannot blind the system.

2.2 Enrichment

Every raw recall is processed through an AI enrichment pipeline that normalizes pathogen names across languages ("Listeria monocytogenes", "Listeriose" both resolve to "Listeria monocytogenes"), standardizes country and jurisdiction names, recovers missing fields by reading the source page directly, assigns severity tiers, and flags outbreak events. Where the regulator publishes category-restricted brand information (e.g. RappelConso's "sans marque" loose retail products), FSIS reads the complementary fields to recover the recalling firm rather than dropping the row as incomplete.

2.3 Verification — dual-reviewer gate

Before any recall reaches the published dataset it must pass two independent AI reviewers, in two passes per day:

- **Morning pass.** First reviewer fetches the regulator's actual page, cross-checks Company / Brand / Product / Pathogen / Date, and proposes corrections. Second reviewer (different model, different vendor) independently verifies. Only rows that both reviewers approve are promoted to the published dataset.
- **Evening pass.** Same two-stage review with a different second-reviewer model — a deliberate dissent check. Catches errors the first pair missed.
- **Both reject → archived.** A row that both reviewers reject is moved to an audit archive, not silently deleted. AFTS retains the full rejection rationale (who, why, when) so the curation history is auditable. This is how FSIS demonstrates the dataset is curated, not just scraped.

A separate URL gate confirms each recall links to a genuine regulatory source (not a news mirror, an aggregator, or a cached snapshot). Recalls that cannot be traced to an official notice are held in a pending queue and re-checked automatically.

2.4 Analysis & delivery

The validated dataset is analyzed at three cadences:

- **Daily.** Yesterday's recalls compiled into a brief with Tier-1 / outbreak triage and regional breakdown.
- **Weekly** (Friday). AI-generated intelligence narrative covering the week's top threats, pathogen distribution, and regional patterns.
- **Monthly** (1st of month). Full 9-section report running descriptive analytics (MoM trends, hotspot detection, cluster analysis, concentration metrics) and active predictive models. The AI narrative interprets every finding through food process engineering principles.

Reports are delivered automatically via email (HTML + PDF attachment for monthly) and are also available on the live dashboard. Custom alert matches trigger immediate email notification. All delivery uses Google Apps Script with deduplication (you never receive the same report twice), staleness checks (stale data is never sent), and one-click unsubscribe.

3. Daily briefs

A daily brief is generated every morning at 10:00 Athens time, covering the previous day's regulatory activity. It is a single HTML page showing:

- **KPI strip** — total recalls, Tier-1 critical count, outbreak count for the day.
- **Regional breakdown** — recalls grouped by region (Europe, North America, Latin America, Asia-Pacific, Middle East & Africa, Oceania).
- **Individual recall cards** — each showing the regulatory source, company name, product description, identified pathogen / hazard, and a direct link to the originating regulatory notice.
- **Tier classification** — every recall is labeled Tier 1 (immediate public health risk) or Tier 2 (significant but lower severity), based on the pathogen's clinical severity and the recall's outbreak status.

Daily briefs are accessible from the dashboard's *Daily* tab and are retained in a rolling archive.

4. Weekly intelligence digest

Every Friday, subscribers receive an email briefing covering the week's global pathogen recall activity. The digest includes:

- **KPI summary** — total recalls, Tier-1 count, outbreak count, and change vs prior week.
- **Leading pathogen card** — the dominant pathogen of the week with case count and percentage share.
- **AI intelligence summary** — a narrative paragraph interpreting the week's data through a food safety engineering lens — naming specific failure modes, production-system risks, and the single most important signal for QA attention.
- **Top 5 critical threats table** — severity-ranked incidents with pathogen, company, product, country, tier / outbreak badges, and direct links to regulatory notices.
- **Full HTML report link** — the complete week's report with all recalls, pathogen distribution, and geographic breakdown.

Each weekly digest is reviewed by AFTS before publication. The Thursday-evening review checkpoint (17:00 Athens) verifies the analytical claims match the underlying data and that the engineering interpretation is sound for the recalls covered. Publication on Friday only proceeds after this review.

5. Monthly intelligence report

The monthly report is the flagship intelligence product — a 9-section analyst-grade briefing produced on the 1st of each month, covering the full prior calendar month. Subscribers receive it as an HTML email with a PDF attachment.

5.1 Intelligence narrative (§ 01)

Three paragraphs of AI-generated analysis, structured as:

- **1. Month headline** — MoM direction, anomaly status, dominant pathogen share, most significant country × pathogen hotspot, composite severity score.
- **2. Structural interpretation** — why the month looks the way it does. Identifies whether the signal is a single-country event, a multi-jurisdictional pattern, or an agency-concentrated artifact. Ties the dominant pathogen to a specific failure mode (environmental harborage, cold-chain breach, thermal underprocess, post-process recontamination).
- **3. Forward-looking recommendation** — the single highest-leverage control a QA director should verify this month, grounded in the predictive model outputs.

When thermal-processing hazards appear in the month's data (e.g. *C. botulinum* in low-acid canned foods), a **Process Authority Note** is automatically appended, speaking in the specialized language of scheduled-process validation.

5.2 Descriptive analytics (§ 02–§ 06)

Month-over-month trend

The current month's recall count is compared against a rolling 3–6 month baseline using a Z-score. $Z > 2$ flags the month as anomalously high; $Z < -2$ flags it as anomalously low. A sparkline chart visualizes the historical trajectory with directional indicators.

Country × Pathogen hotspot matrix

A heatmap of observed recall counts by country and pathogen. Each cell is tested against the expected count under a statistical independence assumption. Cells where the observed count exceeds the expected by more than two standard deviations are flagged as hotspots — indicating a country-pathogen combination occurring more often than chance alone would predict.

Outbreak timeline & cluster analysis

A calendar-axis timeline showing outbreak events with pathogen-coded markers. A sliding 14-day window detects clusters: three or more same-pathogen outbreak events within any two-week span are flagged as a potential coordinated signal.

Regulatory concentration

Two metrics quantify signal structure. The **Herfindahl-Hirschman Index (HHI)** measures how concentrated the recall signal is among regulatory sources — a high HHI may indicate a single-agency activity spike rather than a genuine multi-jurisdictional event. The **Gini coefficient** measures geographic distribution inequality, helping distinguish between country-specific outbreaks and broad international patterns.

Emerging & declining pathogens

Each pathogen's current-month share is compared to its historical average. Pathogens whose share has grown by more than two standard deviations are flagged as emerging; those declining by more than two standard deviations are flagged as receding.

5.3 Predictive outlook (§ 07)

The predictive section runs models progressively as the dataset's history grows. Some models need 12+ months of data to be statistically meaningful; the Predictive Outlook panel shows each model's status, the activation threshold, and (where active) the output. We commit only to what is active today, and disclose what unlocks as the dataset matures.

Status	Model	What it tells you
Active	Linear trend (OLS)	Is overall recall volume trending up, down, or flat? Next-month point forecast with 95% confidence interval.
Active	Poisson forecast	For rare pathogens (e.g. <i>C. botulinum</i>): should we be worried about this month's count? Provides p90 and p95 upper bounds. Observed > p95 is a genuine anomaly.
Active	CUSUM detection	Has something structurally changed in the recall landscape, or is this just month-to-month noise? Detects sustained shifts that single-month comparisons would miss.
Pending (12 mo)	Seasonal decomposition (STL)	Separates trend, seasonality, and residual. Activates at 12 months of history — needs at least one full annual cycle.
Pending (12 mo)	OLS with seasonal dummies	Baseline trend forecast adjusted for seasonal effects.
Pending (24 mo)	Holt-Winters smoothing	Exponential smoothing with trend + seasonality. Activates at 24 months.
Pending (24 mo)	SARIMA	Seasonal autoregressive integrated moving average. The long-form forecast model.
Pending (24 mo)	Prophet	Robust trend + seasonality + holiday effects. Useful for datasets with structural breaks or missing windows.

5.4 Top 10 & Appendix (§ 08 + A)

The Top 10 table ranks the month's most critical incidents by a multi-factor severity sort: outbreak status → *C. botulinum* priority → Tier-1 classification → pathogen severity score → date. The subscriber edition includes **Appendix A**: a complete register of every recall recorded in the month, in the same ranked format.

6. Yearly intelligence report

At the close of each year, FSIS produces a long-range view of global pathogen recall activity. The yearly report is shipped to subscribers as an HTML email with a PDF attachment.

- **Year-over-year trend analysis** — monthly recall volumes across the full available history, with annual comparisons and visual sparklines. As the dataset grows, multi-year patterns and long-range trajectories become visible.
- **Annual pathogen breakdown** — distribution shifts between years: which pathogens are growing, which are receding, which are new entrants.
- **Geographic shift detection** — countries and regions gaining or losing recall share year-over-year.
- **Regulatory agency activity** — which agencies increased or decreased publication frequency, and whether that reflects genuine food safety changes or agency-level operational factors.
- **Forward-looking forecast** — projected recall volumes, seasonal patterns, and emerging-threat candidates for the year ahead, based on all active predictive models.

7. Custom alerts

Subscribers can create personalized watch rules that trigger instant email notifications when a matching recall appears. Each rule combines up to two criteria from four dimensions:

- **Pathogen.** Listeria, Salmonella, E. coli / STEC, C. botulinum, and 20+ others.
- **Product category.** Dairy, meat, seafood, fresh produce, infant formula, beverages, and 30+ categories following Codex Alimentarius classification.
- **Country / jurisdiction.** Any of the 60+ monitored countries.
- **Brand / firm.** Free-text matching against company and brand name fields.

Two criteria can be combined with AND logic ("Listeria in France") or OR logic ("Listeria or Salmonella anywhere"). The alert matcher runs continuously against newly verified recalls. Matches are emailed immediately with a formatted table and direct links to the regulatory notice. Each alert email includes a one-click manage / unsubscribe link.

8. Analytical methods summary

All statistics are computed deterministically from the validated Recalls dataset. The AI narrative receives these pre-computed signals as authoritative context and interprets them — it does not recompute or second-guess them.

Method	Technique	What it measures
MoM Z-score	$(\text{current} - \text{rolling_mean}) / \text{rolling_std}$	Is this month's volume normal or anomalous vs the recent baseline?
Hotspot residuals	Pearson chi-square standardized residuals on the country $\times$ pathogen contingency table	Which country-pathogen combinations are over-represented vs independence?
Cluster detection	14-day sliding window, ≥ 3 same-pathogen events	Are outbreak events sporadic or temporally linked?
Source HHI	Herfindahl-Hirschman Index on source shares	Is the signal coming from many agencies or dominated by one?
Geographic Gini	Gini coefficient on country recall distribution	Are recalls spread globally or concentrated in one region?
Emerging pathogens	Z-score on current-month share vs historical mean share	Which pathogens are growing faster than historical norms?
Severity index	Weighted composite (volume, Tier-1 share, outbreaks, hotspots)	Single 0–100 score summarizing the month's overall threat level.
Linear trend	OLS regression on monthly counts	Next-month forecast with 95% CI and slope direction.
Poisson forecast	Poisson CDF on per-pathogen monthly counts	Upper bounds (p90, p95) for rare pathogens. "Is 3 <i>C. botulinum</i> recalls abnormal?"
CUSUM	Page (1954) tabular cumulative sum	Has the mean recall rate shifted structurally?

9. FSIS in the food safety intelligence landscape

FSIS operates alongside several established platforms that serve the broader food safety and regulatory intelligence space. Each platform has distinct strengths that serve different needs. We see them as fellow contributors to a shared goal: making the global food supply safer through better information.

Platform	Focus	Strengths	FSIS complements by...
RASFF (EU)	EU/EEA regulatory alerts	The authoritative source for EU food and feed safety alerts. Comprehensive, official, trusted by regulators worldwide.	Aggregating RASFF with 60+ non-EU sources into one global view. Many FSIS users monitor RASFF directly — FSIS adds the rest of the world.
HorizonScan (Fera Science)	Global hazard monitoring	Broad hazard coverage beyond pathogens (pesticides, contaminants, fraud). Backed by UK government science.	Offering deeper pathogen-specific analysis: AI narrative with engineering interpretation, predictive models, and process-authority voice.
FoodAKAI	EU food fraud + safety prediction	Pioneering ML-based prediction of food fraud and safety events using RASFF data. Strong academic foundation.	Extending geographic scope beyond the EU, and focusing on pathogen-specific engineering interpretation rather than fraud prediction.
SGS Digicomply	Enterprise regulatory intelligence	Full-scale compliance platform with regulatory change tracking, supplier management, audit workflows. Designed for enterprise food safety teams.	Serving the market SGS does not target: individual QA managers, small-to-mid food companies, consultants, and process authorities who need intelligence without enterprise infrastructure.
FDA iRES / USDA FSIS	US regulatory feeds	The definitive source for US recalls. OpenFDA API enables programmatic access. USDA FSIS provides RSS and email alerts.	Adding 60+ non-US sources alongside FDA / USDA data, plus AI narrative that interprets US recalls in a global context.
INFOSAN (WHO)	International outbreak coordination	The gold standard for international food safety event coordination between national authorities.	Operating in the public-data layer. INFOSAN coordinates between governments; FSIS serves the private sector by monitoring what governments publish. Complementary roles.

We believe more monitoring makes everyone safer. FSIS adds a pathogen-focused, AI-interpreted, process-authority-voiced layer to a landscape where each platform has earned its place.

10. Quality assurance

FSIS commits to a higher standard of dataset quality than raw aggregators by treating verification as a first-class stage of the pipeline, not an after-the-fact spot check. Every recall undergoes the following before it can reach the published dataset:

Source verification. The URL is fetched and confirmed to be a regulator-issued page (not a news mirror, an aggregator, or a cached snapshot). News-mirror domains and known landing-page artifacts are rejected before any reviewer sees the row.

Dual-reviewer approval. Two independent AI reviewers (different models, different vendors, different prompts) must both approve before a row is published. The intent is to catch single-model errors and to surface disagreement when it occurs.

Field-level enrichment. When a regulator publishes a recall with missing or category-restricted fields (e.g. RappelConso "sans marque" loose retail products), reviewers read the source page directly and recover the recalling firm, the affected product, and the implicated pathogen rather than dropping the row.

Audit archive. Rows that both reviewers reject are moved to a Rejected archive — never silently deleted. AFTS retains the full rejection rationale (who rejected, why, when) so the curation history is auditable. This supports the position that the FSIS dataset is curated, not just scraped.

Operator review checkpoint. Every Thursday at 17:00 Athens, AFTS receives a manual-review email summarizing the week's promotions. The Friday weekly digest only ships after this review. This human-in-the-loop step is the central pillar of the FSIS quality model.

These mechanisms are why FSIS subscribers receive a smaller set of recalls than a raw scrape would produce — and why the recalls they do receive are more reliably correct.

11. Data sources

Regulatory agencies

FDA (US), USDA FSIS (US), EU RASFF, FSA (UK), FSANZ (Australia / New Zealand), CFIA (Canada), RappelConso (France), BVL (Germany), AESAN (Spain), EFET (Greece), MPI (New Zealand), SFA (Singapore), CFS (Hong Kong), and 55+ additional national authorities across Europe, Asia-Pacific, Latin America, the Middle East, and Africa. Source coverage depth varies by jurisdiction — some regulators publish via well-structured APIs, others via HTML pages that require ongoing scraper maintenance. AFTS monitors source health continuously and prioritizes the highest-volume jurisdictions for rapid recovery when a regulator changes their publication format.

Pathogen scope

Salmonella, Listeria, E. coli / STEC, C. botulinum, Norovirus, Hepatitis A, Campylobacter, Vibrio, Cyclospora, Yersinia, Bacillus cereus (and cereulide), Brucella, Cronobacter, Staphylococcus aureus, aflatoxin, ochratoxin A, patulin, histamine / scombrototoxin, marine biotoxins, mycotoxins, physical hazards (glass, metal, plastic fragments), chemical contaminants (rodenticides, heavy metals), and pest contamination.

Exclusions

Allergen-only recalls, labeling errors, quality issues (taste, texture, appearance), drug contamination, and product tampering are excluded from the FSIS scope. These are important food safety categories but fall outside the pathogen surveillance mission.

Process authority

Analytical frameworks, severity rubrics, pathogen classification, and the engineering interpretation of each recall are developed under the process authority of AFTS, drawing on in-house expertise in food process engineering, thermal processing, and regulatory compliance. Every view is grounded in validated process engineering: thermal processing (21 CFR 113/114), pasteurization (PMO), aseptic and UHT, hold-tube and F-value lethality, and HACCP.

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